

DSAA3073: Theories in Data Science

Week 1 Validation Sheet · From Uncertainty to Geometry

Coverage: Week 1

Duration: 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Markov's inequality can be applied directly to which variable?

- (a) Any variable with mean zero.
- (b) Any bounded variable.
- (c) A nonnegative variable with a finite mean.
- (d) Only a Gaussian variable.

Solution. (c). Markov's inequality states that if $X \geq 0$ almost surely, $E[X] < \infty$, and $a > 0$, then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Thus a nonnegative random variable with a finite mean is exactly the type of variable to which the inequality applies directly. Boundedness alone does not guarantee nonnegativity, and neither a Gaussian assumption nor a mean-zero assumption is required.

Marking. 5 marks for (c), supported by the nonnegativity and finite-mean conditions; 0 otherwise.

A2 [5 marks]. For fixed confidence level δ , the Week 1 Hoeffding sample-size formula is proportional to $\frac{1}{\varepsilon^2}$. If ε is halved, the sufficient sample size is multiplied by

- (a) 2.
- (b) 4.
- (c) $\frac{1}{2}$.
- (d) $\frac{1}{4}$.

Solution. (b). With δ fixed, write the sufficient sample size as $n = \frac{C}{\varepsilon^2}$, where C contains the unchanged confidence term. Replacing ε by $\frac{\varepsilon}{2}$ gives

$$\frac{C}{\left(\frac{\varepsilon}{2}\right)^2} = \frac{C}{\frac{\varepsilon^2}{4}} = 4 \frac{C}{\varepsilon^2} = 4n.$$

Therefore halving the tolerance requires four times the sufficient sample size under this formula.

Marking. 3 marks for (b); 2 for the inverse-square scaling calculation.

A3 [5 marks]. For $x = (-2, 1, 2)$, which pair is $(\|x\|_2, \|x\|_\infty)$?

- (a) (5, 2)
- (b) (3, 2)
- (c) (3, 1)
- (d) (9, 2)

Solution. (b). By definition,

$$\|x\|_2 = \sqrt{(-2)^2 + 1^2 + 2^2} = \sqrt{4 + 1 + 4} = 3,$$

while

$$\|x\|_\infty = \max(|-2|, |1|, |2|) = 2.$$

Hence the ordered pair is $(3, 2)$.

Marking. 2 marks for the Euclidean-norm calculation; 2 for the maximum-coordinate calculation; 1 for (b).

A4 [5 marks]. Which statement distinguishes a Hilbert space from an arbitrary inner-product space?

- (a) Every pair of vectors is orthogonal.
- (b) Every Cauchy sequence in the induced norm converges to an element of the space.
- (c) The space must have finitely many coordinates.
- (d) Every vector has norm one.

Solution. (b). A Hilbert space is an inner-product space that is complete in the norm induced by its inner product. Completeness means that every Cauchy sequence in that norm converges in norm to an element that still belongs to the space. It does not require finite dimension, unit-length vectors, or pairwise orthogonality. Therefore (b) is the distinguishing statement.

Marking. 3 marks for (b); 2 for identifying completeness in the induced norm.

Part B: Definition in use

B1 [20 marks]. Independent experimental units produce identically distributed pairs (A_i, B_i) , $i = 1, \dots, n$. Different pairs are independent, but A_i and B_i within the same pair may be dependent. Each coordinate lies in $[0, 1]$. The target is $\Delta = E[A_i - B_i]$, estimated by

$$\hat{\Delta}_n = \frac{1}{n} \sum_{i=1}^n (A_i - B_i).$$

Derive a valid two-sided exponential guarantee for $|\hat{\Delta}_n - \Delta|$ and a sufficient sample size for confidence $1 - \delta$. Your reasoning must make clear why the possible within-pair dependence does not invalidate the guarantee you use.

Solution. The estimator is an average of differences, so the useful independent summand is one complete experimental unit:

$$Y_i = A_i - B_i.$$

This choice matters because the prompt guarantees independence between different pairs, not between the two coordinates inside one pair. Each Y_i is a function only of pair i ; therefore $Y_1, \dots, Y_n$ are independent across i . No within-pair independence is needed for this conclusion.

Linearity of expectation, which does not require independence within a pair, gives

$$E[Y_i] = E[A_i - B_i] = \Delta.$$

Since $A_i, B_i \in [0, 1]$, the smallest possible difference is $0 - 1 = -1$ and the largest is $1 - 0 = 1$. Hence

$$-1 \leq Y_i \leq 1,$$

so every interval width is $b_i - a_i = 2$. Also

$$\hat{\Delta}_n - \Delta = \frac{1}{n} \sum_{i=1}^n (Y_i - E[Y_i]).$$

Hoeffding's two-sided bounded-sum inequality applies because the Y_i are independent and have known bounded ranges. The event $|\hat{\Delta}_n - \Delta| \geq \varepsilon$ is equivalent to $|\sum_i (Y_i - \Delta)| \geq n\varepsilon$. Substituting $t = n\varepsilon$ and $\sum_i (b_i - a_i)^2 = 4n$ gives

$$P(|\hat{\Delta}_n - \Delta| \geq \varepsilon) \leq 2 \exp\left(-2n^2 \frac{\varepsilon^2}{4n}\right) = 2 \exp\left(-n \frac{\varepsilon^2}{2}\right).$$

To make this probability at most δ , require

$$2 \exp\left(-n \frac{\varepsilon^2}{2}\right) \leq \delta.$$

Taking natural logarithms and rearranging yields

$$-n \frac{\varepsilon^2}{2} \leq \ln\left(\frac{\delta}{2}\right) \rightarrow n \geq 2 \frac{\ln\left(\frac{2}{\delta}\right)}{\varepsilon^2}.$$

Because a sample size must be an integer, an explicit sufficient choice is $n = \left\lceil 2 \frac{\ln\left(\frac{2}{\delta}\right)}{\varepsilon^2} \right\rceil$. Therefore the displayed exponential bound is valid despite possible dependence within a pair: the independent observations used by Hoeffding are the paired differences Y_i , one per experimental unit.

Marking. 6 marks for defining $Y_i = A_i - B_i$ and transferring across-pair independence; 4 for deriving its mean and range $[-1, 1]$; 5 for the two-sided Hoeffding substitution and exponent; 3 for solving the confidence inequality and giving a sufficient integer sample size; 2 for explicitly explaining why within-pair dependence is harmless. Preserve method credit for a correct paired-difference route after arithmetic errors, and follow through consistently when solving for n .

Part C: Basic skills

C1 [7 marks]. A nonnegative random variable satisfies $E[X] = 4$. Use Markov's inequality to bound $P(X \geq 20)$.

Solution. Markov's inequality applies because X is nonnegative, has finite mean, and the threshold 20 is positive. It gives

$$P(X \geq 20) \leq \frac{E[X]}{20} = \frac{4}{20} = \frac{1}{5}.$$

Thus Markov certifies the upper bound $P(X \geq 20) \leq \frac{1}{5}$; it does not claim that the probability equals $\frac{1}{5}$.

Marking. 3 marks for selecting Markov and checking its conditions; 2 for substituting $E[X] = 4$ and threshold 20; 2 for the bound $\frac{1}{5}$ with the inequality direction retained.

C2 [8 marks]. For $u = (1, 2, -2)$ and $v = (2, -1, 2)$, compute $\langle u, v \rangle$, $\|u\|_2$, and $\|v\|_2$.

Solution. Using the Euclidean inner-product definition,

$$\langle u, v \rangle = (1)(2) + (2)(-1) + (-2)(2) = 2 - 2 - 4 = -4.$$

Using $\|z\|_2 = \sqrt{\sum_j z_j^2}$,

$$\|u\|_2 = \sqrt{1^2 + 2^2 + (-2)^2} = \sqrt{9} = 3,$$

and

$$\|v\|_2 = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3.$$

Therefore the requested values are -4 , 3 , and 3 , respectively.

Marking. 4 marks for the inner-product definition, substitution, and value; 2 marks for each norm calculation. Allow follow-through credit for correct definitions with a single arithmetic slip.

Part D: Construction and counterexample

D1 [17 marks]. Construct a nonzero vector $y = (a, b, c)$ that is orthogonal to both $u = (1, 1, 0)$ and $v = (0, 1, 1)$. Verify both inner products. Then decide whether either inequality $|\langle y, u \rangle| \leq \|y\|_2 \|u\|_2$ or $|\langle y, v \rangle| \leq \|y\|_2 \|v\|_2$ is an equality, and justify your decision.

Solution. A vector is orthogonal to another vector exactly when their inner product is zero. Thus the two requirements become

$$\langle (a, b, c), (1, 1, 0) \rangle = a + b = 0, \quad \langle (a, b, c), (0, 1, 1) \rangle = b + c = 0.$$

These equations give $a = -b$ and $c = -b$. Choosing any nonzero value of b produces a nonzero solution. For example, taking $b = -1$ gives

$$y = (1, -1, 1).$$

Direct verification yields $\langle y, u \rangle = 1 - 1 = 0$ and $\langle y, v \rangle = -1 + 1 = 0$.

For nonzero vectors, equality in Cauchy–Schwarz holds exactly when the two vectors are linearly dependent. Here y is not a scalar multiple of u : the third coordinate of every multiple of u is zero, whereas $y_3 = 1$. It is also not a scalar multiple of v : every multiple of v has first coordinate zero, whereas $y_1 = 1$. Hence the right sides $\|y\|_2\|u\|_2$ and $\|y\|_2\|v\|_2$ are positive, while both left sides are zero. Both inequalities are therefore strict, not equality cases.

Marking. 5 marks for the two orthogonality equations; 4 for any valid nonzero solution; 4 for verifying both inner products; 4 for applying the nonzero equality condition and proving non-dependence with each target vector. Give method credit for a correct one-parameter family and follow through with the learner's valid nonzero choice.

D2 [18 marks]. Define $q : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$q(x) = |x_1 + x_2| + |x_1 - x_2|.$$

Decide whether q is a norm. Give a complete justification of your decision; checking a few numerical vectors is not sufficient.

Solution. The function is a norm. This requires verification for arbitrary vectors and scalars, rather than numerical tests.

Nonnegativity and definiteness. Each absolute value is nonnegative, so $q(x) \geq 0$. If $q(x) = 0$, then a sum of two nonnegative terms is zero; hence both terms are zero:

$$x_1 + x_2 = 0, \quad x_1 - x_2 = 0.$$

Adding the equations gives $2x_1 = 0$, and subtracting them gives $2x_2 = 0$. Thus $x = (0, 0)$. Conversely, substituting $x = 0$ gives $q(0) = 0$. Therefore $q(x) = 0$ exactly when $x = 0$.

Absolute homogeneity. For every scalar c ,

$$q(cx) = |cx_1 + cx_2| + |cx_1 - cx_2| = |c|(|x_1 + x_2| + |x_1 - x_2|) = |c|q(x).$$

Triangle inequality. Let arbitrary $x, y \in \mathbb{R}^2$ be given. Applying the scalar triangle inequality separately to the two linear forms gives

$$q(x + y) = |(x_1 + x_2) + (y_1 + y_2)| + |(x_1 - x_2) + (y_1 - y_2)|$$

and hence

$$q(x + y) \leq |x_1 + x_2| + |y_1 + y_2| + |x_1 - x_2| + |y_1 - y_2| = q(x) + q(y).$$

All four norm requirements hold for every admissible input, so q is a norm on $\mathbb{R}^2$.

Marking. 2 marks for the correct classification; 5 for nonnegativity and the two directions of definiteness; 3 for absolute homogeneity for arbitrary scalars; 6 for the general triangle-inequality derivation; 2 for explicitly concluding that all norm axioms hold. Preserve method credit when a valid norm-axiom audit contains a minor algebra slip.

Part E: Transfer through a new representation

E1 [25 marks]. Two independent instruments estimate the same mean μ . Instrument A supplies n independent readings $A_i \in [0, 1]$; instrument B supplies n independent readings $B_i \in [-1, 1]$. All $2n$ readings are mutually independent. For $0 \leq \alpha \leq 1$, consider

$$\hat{\mu}_\alpha = \alpha \left(\frac{1}{n} \sum_i A_i \right) + (1 - \alpha) \left(\frac{1}{n} \sum_i B_i \right).$$

Among all choices of α , determine the one giving the strongest two-sided exponential guarantee obtainable from the Week 1 assumptions for $|\hat{\mu}_\alpha - \mu|$. Give that guarantee and prove that no other $\alpha \in [0, 1]$ gives a better exponent by this route.

Solution. The two instruments have different ranges, so treating the estimator as an ordinary equal-width average would discard information needed to compare the choices of α . Instead, write it as a sum of its $2n$ independent weighted summands:

$$\hat{\mu}_\alpha = \sum_{i=1}^n \frac{\alpha}{n} A_i + \sum_{i=1}^n \frac{1-\alpha}{n} B_i.$$

Mutual independence of all readings implies independence of these weighted summands. Because both instruments estimate the same mean,

$$E[\hat{\mu}_\alpha] = \alpha\mu + (1 - \alpha)\mu = \mu,$$

so the estimator is unbiased for every $\alpha \in [0, 1]$.

Multiplying an interval by a nonnegative weight multiplies its width by that weight. Each A term starts with width 1 and therefore has weighted width $\frac{\alpha}{n}$. Each B term starts with width 2 and therefore has weighted width $2\frac{1-\alpha}{n}$. Hoeffding's denominator is the sum of the squared widths of all independent summands:

$$n \left(\frac{\alpha}{n} \right)^2 + n \left(2 \frac{1-\alpha}{n} \right)^2 = \frac{\{\alpha^2 + 4(1-\alpha)^2\}}{n}.$$

Applying the two-sided Hoeffding bound to deviations of the full weighted sum from its mean gives, for $\varepsilon > 0$,

$$P(|\hat{\mu}_\alpha - \mu| \geq \varepsilon) \leq 2 \exp \left(-2n \frac{\varepsilon^2}{\alpha^2 + 4(1-\alpha)^2} \right).$$

For fixed n and ε , a stronger exponential guarantee has a more negative exponent. Since the numerator is fixed and the denominator is positive, this means minimizing

$$d(\alpha) = \alpha^2 + 4(1-\alpha)^2.$$

Expanding and completing the square gives

$$d(\alpha) = 5\alpha^2 - 8\alpha + 4 = 5 \left(\alpha - \frac{4}{5} \right)^2 + \frac{4}{5}.$$

The squared term is nonnegative and vanishes exactly at $\alpha = \frac{4}{5}$, which lies in the allowed interval $[0, 1]$. Hence this is the unique global minimizer, and the minimum denominator is $\frac{4}{5}$. Substitution yields

$$P(|\hat{\mu}_{\frac{4}{5}} - \mu| \geq \varepsilon) \leq 2 \exp \left(-2n \frac{\varepsilon^2}{\frac{4}{5}} \right) = 2 \exp \left(-\left(\frac{5}{2} \right) n \varepsilon^2 \right).$$

For any other $\alpha \in [0, 1]$, the square is strictly positive, so $d(\alpha) > \frac{4}{5}$ and the exponent has smaller magnitude. Therefore no other admissible weight provides a stronger Hoeffding exponent by this route.

Marking. 6 marks for the weighted-summand representation, independence, and unbiasedness; 5 for deriving both weighted interval widths and their squared sum; 4 for the two-sided bound as a function of α ; 6 for choosing the denominator as the optimization target and proving the unique global minimizer by the square decomposition; 4 for the final guarantee and the no-better-exponent

conclusion. Preserve method and follow-through credit after algebra errors in a valid weighted-width setup. Mere substitution into an equal-width sample-mean formula earns at most 5 marks.

Part F: Comparison and limitation

F1 [15 marks]. A lab records one Bernoulli variable U with $P(U = 1) = P(U = 0) = \frac{1}{2}$, then enters the same observed value into its spreadsheet n times: $X_1 = \dots = X_n = U$. An analyst argues that the average of these bounded entries has deviation probability at most $2 \exp(-2n\varepsilon^2)$.

Audit the claim. Give explicit n and ε for which the claimed inequality is false, and identify precisely what the spreadsheet failed to create despite containing n bounded entries.

Solution. Because every spreadsheet entry equals the same draw U , its average is

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} (nU) = U.$$

Also $E[\hat{\mu}_n] = E[U] = \frac{1}{2}$. Since U is either 0 or 1,

$$\left| \hat{\mu}_n - \frac{1}{2} \right| = \left| U - \frac{1}{2} \right| = \frac{1}{2}$$

on both possible outcomes. Consequently, for every $0 < \varepsilon \leq \frac{1}{2}$,

$$P\left(\left| \hat{\mu}_n - \frac{1}{2} \right| \geq \varepsilon\right) = 1.$$

To contradict the claimed bound explicitly, take $\varepsilon = 0.4$ and $n = 10$. The claimed right-hand side is

$$2 \exp(-2(10)(0.4)^2) = 2e^{-3.2} \approx 0.0815 < 1,$$

whereas the actual deviation probability is 1. Thus the proposed inequality would assert $1 \leq 0.0815$, which is false.

The Hoeffding guarantee for n entries requires n independent random observations. Copying one bounded draw into n cells preserves boundedness but does not create independence or additional randomness: $X_1, \dots, X_n$ are perfectly dependent and contain only one random draw. Therefore the analyst has applied an n -independent-observation guarantee to a dataset with only one independent unit.

Marking. 4 marks for deriving $\hat{\mu}_n = U$; 4 for proving the deviation is $\frac{1}{2}$ with probability one; 4 for any valid explicit (n, ε) making the claimed right side below one and comparing both sides; 3 for identifying the missing independent observations and perfect dependence. Give follow-through credit for a valid contradiction using different numerical values.